

TIANZHI XIANG

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🔍 RESEARCH INTERESTS

Acoustic Metasurfaces • Wave Mechanics and Control • Acoustic–Elastic Wave Coupling
Acoustic Communication • Interfacial Wave Transmission

🎓 EDUCATION

Tianjin University, School of Mechanical Engineering, Tianjin, China Sep. 2024 – Present

- *M.S. Candidate in Mechanics* | Expected Jun. 2027
- **Core courses:** Mechanical Metamaterials (99/100), Foundation of Solid Mechanics (93/100).

IMDEA Materials Institute, Madrid, Spain Aug. 2026 – Present

- *China Scholarship Council (CSC)* funded Visiting *M.S. Candidate*.
- **Host Supervisor:** Prof. Johan Christensen.

Tianjin University, School of Mechanical Engineering, Tianjin, China Sep. 2020 – Jun. 2024

- *B.S. in Engineering Mechanics* (National Strengthening Foundation Plan), GPA: 85/100.
- **Core courses:** Theoretical Mechanics (94/100), Advanced Elasticity (96/100), Vibration and Measurement (90/100), Ordinary Differential Equations and Motion Stability (93/100).

💡 RESEARCH HIGHLIGHTS

- First-author manuscript under review on vibro-acoustic metasurfaces for efficient airborne sound transmission through solid barriers, addressing wave coupling across strongly impedance-mismatched interfaces.
- Developed scattering-matrix and interfacial-matching models to describe acoustic–elastic wave transfer, providing a basis for understanding cross-interface coupling and air-coupled elastic-wave excitation.
- Demonstrated metasurface-based platforms for wavefront manipulation, tunable asymmetric transport, and non-contact vibro-acoustic excitation, with a broader interest in extending passive wave-control strategies toward Willis-coupled, programmable, or active acoustic and elastic-wave systems.

⚙️ RESEARCH EXPERIENCE

[Ongoing] Direction-dependent dual-mode conversion using air-coupled metasurfaces Apr. 2026 – Present

Advisor: Prof. Yanfeng Wang & Prof. Johan Christensen

- Developing metasurfaces for multi-mode conversion between airborne sound and elastic wave
- Formulating Floquet–Willis, impedance-based, and scattering-matrix models to clarify momentum matching, modal coupling, and air-solid energy transfer mechanisms.
- Designing asymmetric metasurface structures and conducting numerical and experimental validation, aiming to enhance mode-conversion efficiency, directional selectivity, and operating bandwidth.

Vibro-acoustic metasurfaces for sound transmission enhancement through air–solid interfaces Sep. 2024 – Mar. 2026

Advisor: Prof. Yanfeng Wang

- Developed scattering-matrix and interfacial-matching models to explain airborne-sound-to-elastic-wave conversion across air–solid interfaces.
- Identified metasurface-induced interfacial matching as the key mechanism for overcoming weak sound–structure coupling and enabling thickness-insensitive transmission enhancement.
- Verified high-efficiency airborne sound penetration through solid barriers using acoustic measurements, vibration experiments, and transmission analysis.

Phase-Gradient Acoustic Metasurfaces for Wavefront Manipulation

Jul. 2021 – Jun. 2023

Project Leader | Tianjin Municipal Undergraduate Innovation Training Program

Advisor: Prof. Yuesheng Wang & Prof. Yanfeng Wang

- Designed acoustic metasurfaces for wavefront manipulation based on Generalized Snell’s Law and impedance descriptions.
- Developed parametric unit cell models, forming a basis for later vibro–acoustic interfacial matching research.

 PROJECTS

Reconfigurable Acoustic Moiré Metasurface

Oct. 2024 – Jan. 2025

Role: **Second author.**

- Conducted acoustic experiments and analyzed transmission data to characterize tunable asymmetric acoustic transport in three-dimensional free space.
- Evaluated amplitude contrast, propagation direction, and backward suppression under different relative twist configurations.

RoboMaster University League Competition (DJI)

Sep. 2022 – Jul. 2023

Role: **Team lead.**

- Led a 40-member multidisciplinary team through one year of preparation, coordinating project planning, technical discussions, task allocation, and competition execution.
- Contributed to system-level R&D planning and mechanical design review, including module integration, milestone tracking, structural feasibility, assembly constraints, and reliability evaluation.
- **Achieved First Prize.**

2024 Global Youth Impact Hackathon (WFUNA)

Dec. 2024

Role: Participant.

- Collaborated in a 48-hour multidisciplinary challenge to develop and present a plastic-pollution solution concept under SDG 14, covering feasibility and environmental impact.

 HONORS & AWARDS

First-Class Academic Scholarship for Graduate Students

Sep. 2024

Outstanding Graduate of Tianjin University

Jun. 2024

Merit Student of Tianjin University

Dec. 2023

Outstanding Student Leader Award

Sep. 2021

 PUBLICATIONS & PATENTS

- **T. Z. Xiang** et al., “Vibro-acoustic metasurfaces for thickness-independent penetration of airborne sound through solid barriers.” Under review at *International Journal of Mechanical Sciences*.
- H. T. Zhou, **T. Z. Xiang** et al., “Reconfigurable Asymmetric Acoustic Transport Enabled by a Reciprocal Moiré Metasurface.” Under review at *National Science Review*.
- **T. Z. Xiang** et al., “A Fabrication Method of an Acoustic Metasurface Structure for Efficient Sound Transmission Through Solid Barriers.” *Chinese Invention Patent* CN119446109B, granted, 2025.

 TECHNICAL SKILLS

- **Language** Mandarin, English (IELTS Academic 7.0)
- **Software** COMSOL, SolidWorks, AutoCAD
- **Programming** MATLAB, Python, L^AT_EX
- **Computational Design** Genetic Algorithms, Topology Optimization
- **Experiments** Acoustic Measurement, Laser Vibrometry